

Week 3: Agentic Text-to-SQL System

Final Findings & Execution Report

1. Repository Information

GitHub URL: https://github.com/Zahlencraft/week_3

2. Project Overview

This project successfully implements an AI-powered agentic system capable of understanding natural language questions, decomposing them into structural components, generating and executing safe SQL queries, handling errors via automated retries, and returning meaningful results.

3. Detailed Execution Findings

Task 1: SQL Benchmark Dataset & Evaluation

A comprehensive benchmark dataset of 50 natural language questions was created, each mapped to a verified ground-truth SQL query. An evaluation strategy was formalized, focusing on functional execution success rather than strict string matching, allowing for query variability.

Task 2: Query Understanding (Decomposition)

All 50 queries were systematically decomposed into structural components: Intent, Tables, Columns, Filters, and Joins. This structural breakdown provided the necessary context to guide the LLM effectively during the Text-to-SQL generation phase.

Task 3: Pipeline & Safety Execution

A robust Python-based execution pipeline was built. Critical features include:

- A query validator that prevents destructive operations (e.g., DROP, DELETE, UPDATE).
- An automated retry mechanism that catches execution errors and prompts the LLM to correct the SQL syntax.
- Demonstrated 100% execution success on the benchmark dataset during final testing.

Task 4: Mini SQL Agent API

The full pipeline was encapsulated within a FastAPI web service. It exposes a standardized endpoint that processes questions, orchestrates the Text-to-SQL pipeline, executes the queries safely, and returns synthesized natural language summaries alongside the raw data.

4. Conclusion

The week 3 assignments have been fully implemented. The resulting system is a resilient, self-correcting agent capable of robust database interaction through natural language, laying a strong foundation for future enhancements in Agentic workflows.